

MICRO-PROJECT

Contact Management System



Submitted- 2023-24

This micro-project work submitted in partial fulfillment of requirements for the award of diploma in Computer Technology for subject "**Python Programing**"(22616)".

Computer Engineering

Maharashtra State Board of Technical Education, Mumbai

UNDER THE GUIDANCE OF

Prof.S.K.Kharkate

Lect. In Computer Technology

SUBMITTED BY

Sr no.	Roll no.	Name of Student	Enrollment no.
1.	8	Shreyash Vinayak Bagade	2101210085

DEPARTMENT OF COMPUTER ENGINEERING



GOVERNMENT POLYTECHNIC, BRAMHAPURI DIST CHANDRAPUR

DEPARTMENT OF COMPUTER ENGINEERING

Certificate

This is to certify that the following student of this institute have carried out this micro-project in work “*Contact Management System*” Under the guidance of **Prof.S.K.Kharkate** Lect. in the Department of Computer Engineering during the session 2023-2024 This work has been in the partial fulfillment of the award for Diploma in Computer Engineering from Maharashtra State Board of Technical Education, Mumbai.

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1.	8	Shreyash Vinayak Bagade	2101210085

Project Guide

Head of department

Principal

Prof. S. K. Kharkate

Prof. S. K. Kharkate

Dr.R.L.Wankhade

Part-A

1.0.Aim of micro project:-

Contact Management System

2.0. Brief introduction:

The Contact Management System, built using Python, is an exemplary demonstration of effective file handling techniques. This system provides users with a seamless experience to manage their contact information efficiently. Leveraging the power of file handling in Python, it enables users to add, delete, update, and view contact details stored within a text file. By embracing file handling principles, the Contact Management System ensures data persistence and accessibility, allowing users to seamlessly interact with their contacts across multiple sessions. With a focus on robust file operations, this system offers a reliable solution for organizing and maintaining contact information, demonstrating the practical applications of file handling in real-world scenarios.

3.0.Course outcome:

- Display message on screen using Python script on IDE.
- Develop python program to demonstrate use of Operators
- Perform operations on data structures in Python.
- Develop functions for given problem.
- Design classes for given problem.

4.0.Literature review:

- The Information about the topic was collected from different source such as different sites. As well as using web links. Also we discussed the topic with subject related teacher. Also in lecture our subject teacher gives us information about this project.
- What is Python ?

<https://youtu.be/B7qB62Msems?si=xbaqyVN5Brmfb0Px>

- What is file handling?

<https://youtu.be/VxMV6wGS3NY>

- What is GUI in python?

<https://youtu.be/Z2Lgml>

5.0. Proposed methodology:

Algorithm: Contact Management System

Step1 :- Import Necessary Libraries: Import the os module for file operations.

Step2 :- Define Functions:

- add_contact(name, phone): Adds a new contact to the contacts file.
- delete_contact(name): Deletes a contact from the contacts file.
- update_contact(name, new_phone): Updates the phone number of a contact.
- show_all_contacts(): Displays all contacts stored in the contacts file.
- main(): Main function to drive the contact management system.

Step3 :- Implement Add Contact Functionality:

- Prompts the user to enter the name and phone number of the new contact.
- Appends the contact information to the contacts file.
- Prints a success message.

Step4 :- Implement Delete Contact Functionality:

- Prompts the user to enter the name of the contact to be deleted.
- Reads the contacts file, removes the specified contact, and writes back to the file.
- Prints a success message.

Step5 :- Implement Update Contact Functionality:

- Prompts the user to enter the name of the contact to be updated and the new phone number.
- Reads the contacts file, updates the phone number of the specified contact, and writes back to the file.

- Prints a success message if the contact is found, otherwise prints a message indicating that the contact was not found.

Step6 :- Implement Show All Contacts Functionality:

- Reads the contacts file and prints all contacts with their names and phone numbers.

Step7 :- Implement Main Functionality:

- Displays a menu of options to the user.
- Based on the user's choice, calls the corresponding function.
- Continues running until the user chooses to exit.

Step8 :- Ensure File Existence:

- Checks if the contacts file exists. If not, creates an empty file.

Step9 :- Run the Program:

Executes the main() function to start the contact management system.

7.0 Action plan:

Sr. no.	Detail of Activity	Start Date	Finish Date	Name of student
1	Search for topic	30/1/24	30/1/22	Shreyash v Bagade
2	Information search and resource collected	31/1/24	31/1/24	Shreyash v Bagade
3	Discuss with project guide	31/1/24	1/1/24	Shreyash v Bagade
4	Algorithm	2/1/24	3/1/24	Shreyash v Bagade
5	Implement part-A	4/1/24	5/1/24	Shreyash v Bagade
6	Implement Test cases			Shreyash v Bagade
8	Implement part-B			Shreyash v Bagade
9	Submission of final report			Shreyash v Bagade

Part:-B

“Contact Management System”

1.0. Rationale:

The Contact Management System, built using Python, is an exemplary demonstration of effective file handling techniques. This system provides users with a seamless experience to manage their contact information efficiently. Leveraging the power of file handling in Python, it enables users to add, delete, update, and view contact details stored within a text file. By embracing file handling principles, the Contact Management System ensures data persistence and accessibility, allowing users to seamlessly interact with their contacts across multiple sessions. With a focus on robust file operations, this system offers a reliable solution for organizing and maintaining contact information, demonstrating the practical applications of file handling in real-world scenarios.

2.0. Aim of Micro-project:

Contact Management System

3.0. Course outcomes Addressed:

- Display message on screen using Python script on IDE.
- Develop python program to demonstrate use of Operators
- Perform operations on data structures in Python.
- Develop functions for given problem.
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4.0. Literature review:

- The Information about the topic was collected from different source such as different sites. As well as using web links. Also we discussed the topic with subject related teacher. Also in lecture our subject teacher gives us information about this project.
- What is Software Testing?
<https://youtu.be/B7qB62Msems?si=xbaqyVN5Brmfb0Px>
- What are the types of testing?
<https://youtu.be/VxMV6wGS3NY>

- What are test cases?
<https://youtu.be/Z2LgmI>

5.0. Actual Methodology followed:

- Firstly we will study about given topic.
- Then after we will study and discuss about topic with our project guide.
- After we have collected the information of our topic from internet and Python book, After we have started to working of micro project.
- After we start the making test cases and test the all functionality of MS-word.
- When our micro-project is ready then we submitted to our project guide.

6.0. Resources used:

Name of Resource/Material	Specification
Laptop	SSD 512GB, RAM 8 GB ,Intel i5 processor
Software	MS-Word, MS-Excel, Pycharm ,Chrome
Other Accessories	Reference Book Head First Python Autor:-Paul Barry

7.0.Program :-

```
import os

def add_contact(name, phone):
    with open("contacts.txt", "a") as file:
        file.write(f"{name},{phone}\n")
    print("Contact added successfully.")

def delete_contact(name):
```



```
contacts = []
with open("contacts.txt", "r") as file:
    for line in file:
        contact_name, _ = line.strip().split(",")
        if contact_name.lower() != name.lower():
            contacts.append(line)
with open("contacts.txt", "w") as file:
    file.writelines(contacts)
print("Contact deleted successfully.")
```

```
def update_contact(name, new_phone):
    contacts = []
    updated = False
    with open("contacts.txt", "r") as file:
        for line in file:
            contact_name, phone = line.strip().split(",")
            if contact_name.lower() == name.lower():
                contacts.append(f"{contact_name},{new_phone}\n")
                updated = True
            else:
                contacts.append(line)
    with open("contacts.txt", "w") as file:
        file.writelines(contacts)
    if updated:
        print("Contact updated successfully.")
    else:
        print("Contact not found.")
```

```
def show_all_contacts():
    with open("contacts.txt", "r") as file:
        print("\n===== All Contacts =====")
        for line in file:
            name, phone = line.strip().split(",")
            print(f"Name: {name}, Phone: {phone}")
```

```
def main():
    while True:
        print("\n===== Contact Management System =====")
        print("1. Add Contact")
        print("2. Delete Contact")
        print("3. Update Contact")
        print("4. Show All Contacts")
        print("5. Exit")

        choice = input("Enter your choice: ")

        if choice == "1":
            name = input("Enter contact name: ")
            phone = input("Enter contact phone number: ")
            add_contact(name, phone)
        elif choice == "2":
            name = input("Enter contact name to delete: ")
            delete_contact(name)
        elif choice == "3":
            name = input("Enter contact name to update: ")
            new_phone = input("Enter new phone number: ")
            update_contact(name, new_phone)
        elif choice == "4":
            show_all_contacts()
        elif choice == "5":
            print("Exiting program.")
            break
        else:
            print("Invalid choice. Please try again.")

if __name__ == "__main__":
    if not os.path.exists("contacts.txt"):
        with open("contacts.txt", "w"):
            pass # Create the contacts file if it doesn't exist
    main()
```

8.0.Output:-

```
===== Contact Management System =====
1. Add Contact
2. Delete Contact
3. Update Contact
4. Show All Contacts
5. Exit
Enter your choice: 1
Enter contact name: shreyash
Enter contact phone number: 8010159779
Contact added successfully.
```

```
===== Contact Management System =====
1. Add Contact
2. Delete Contact
3. Update Contact
4. Show All Contacts
5. Exit
Enter your choice: 4
```

```
===== All Contacts =====
Name: shreyash, Phone: 8010159779
```

```
===== Contact Management System =====
1. Add Contact
2. Delete Contact
3. Update Contact
4. Show All Contacts
5. Exit
Enter your choice: 3
Enter contact name to update: shreyash
Enter new phone number: 9765236126
Contact updated successfully.
```

```
===== Contact Management System =====
1. Add Contact
2. Delete Contact
3. Update Contact
4. Show All Contacts
5. Exit
```

```
===== Contact Management System =====
1. Add Contact
2. Delete Contact
3. Update Contact
4. Show All Contacts
5. Exit
Enter your choice: 2
Enter contact name to delete: shreyash
Contact deleted successfully.

===== Contact Management System =====
1. Add Contact
2. Delete Contact
3. Update Contact
4. Show All Contacts
5. Exit
Enter your choice: |
```

9.0. Skill Developed / Learning outcome of this Micro-Project

- Database Management (SQLite)
- CRUD Operations
- Debugging
- Python Programming

10.0. Application of This Project

- Small Business Contact Management
- Personal Contact Organizer
- Event Planning and Coordination

11.0. Reference:

- What is Software Testing?
<https://youtu.be/B7qB62Msems?si=xbaqyVN5Brmfb0Px>
- What are the types of testing?
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